

Abhishek Thakur

Master Student - Mechatronics and Robotics

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Education

Masters in Mechatronics and Robotics

Hochschule Schmalkalden

10.2023 – Present | Schmalkalden, Germany

Bachelors in Mechanical Engineering

*K. K. Wagh Institute of Engineering Education and Research
(affiliate with Savitribai Phule Pune University)*

06.2017 – 05.2021 | Nashik, India

Experience

Project Assistant

Eviotec Technologieentwicklung Prof. Schrödel & Rödel GbR

04.2025 – 08.2025 | Schmalkalden, Germany

Tech stack: STM32, Embedded C, UV & safety sensors, Fusion 360, DialluxEvo

- Developed embedded control firmware for a sensor-driven electromechanical UVC system, integrating UV and safety sensors for reliable closed-loop operation.
- Contributed to system-level design decisions, balancing performance, safety, and cost constraints for industrial deployment.
- Performed modeling, simulation, and rapid prototyping to evaluate design alternatives, reducing operation time from 15 minutes to 5 minutes while maintaining ~95% system efficiency.
- Conducted experimental validation by correlating simulated and measured system behavior (<10% deviation) and documented test procedures to ensure traceability and reproducibility

Hackathon

Assistive robot

Claude Agent SDK, MCP, Robot Manipulation, HRI, NormaCore Station API.

06.2026 | HackLab, Berlin

Developed an agentic robotic assistant that autonomously interprets natural-language voice and chat commands to perform object manipulation tasks on the NormaCore Eirobot. Integrated AI agents, computer vision, and the NormaCore Station API to enable perception, planning, grasping, stacking, delivery, and task verification.

Skills

Programming: C++, Python

Tools: Linux, ROS/ROS2, Git, Docker, Gazebo, MS Office, PLC systems

Prototyping: CAD-Solidworks, 3D printing, KiCAD, Embedded systems

Languages: English: (Fluent), German: B1/B2

Vision/ML tools: Pytorch, OpenCV, ollama

Projects

Remote Robotics Training Platform with CrossLab Integration

Hochschule Schmalkalden – in collaboration with TU Freiberg

04.2025 – Present

Technologies: Docker, XPRA, ROS2, CLI, Git, Teensy

- Developed a ROS 2–based digital twin for real-time co-simulation and remote operation of a 4-DOF robotic arm, enabling closed-loop control and feedback over a networked setup.
- Implemented Jacobian-based inverse kinematics and integrated MoveIt 2 motion planning for model-based control and trajectory execution.
- Designed a modular system architecture combining simulation, embedded control, and remote experimentation; validated functionality through repeated experimental use in an academic setting.

Curriculum-Based Soft Actor-Critic Learning for 7-DOF Robotic Manipulation in Simulation

Tech stack: PyTorch, PyBullet, Reinforcement Learning (SAC)

02.2026 – 03.2026

- Implemented a Soft Actor-Critic framework for continuous control of a 7-DOF KUKA iiwa manipulator in a physics-based simulation environment.
- Designed and evaluated reward shaping strategies and a curriculum learning pipeline (progressive target distributions) to improve sample efficiency and training stability.
- Achieved around 80% success rate within 500 episodes; analyzed convergence behavior, instability modes, and sensitivity to hyperparameters using custom logging and evaluation metrics.

Fruit Sorting Robot

Imitation Learning LeRobot, ACT, AWS, Hugging Face

05.2026 | EPFL, Lausanne, Switzerland

- Collected 150 demonstrations across 3 fruit categories; trained ACT policy on H200 GPU via Hugging Face computeAchieved ~60% task success rate with training cut short at 30k/100k steps due to time constraints; policy showed consistent colour-based discrimination with occasional pick failures

Certificates

e-Series Core Track (Universal Robots)

Python Data Structures (University of Michigan)

Machine Learning (Stanford Online)

Aerial Robotics (University of Pennsylvania)

Design Thinking for Innovation (University of Virginia)

Additional Courses

SUAS Premier

Funded by BMBF & DAAD

09.2024 – 12.2024 | Schmalkalden, Germany

- Completed a BMBF-funded “Profi plus” program focused on technical and professional integration into the German engineering job market
- Executed multiple hands-on mini-projects in embedded systems and ROS-based robotics, covering system integration and basic control workflows
- Applied MINT skills in structured problem-solving, technical documentation, and engineering communication
- Developed workplace competencies aligned with German industry practices, including teamwork and task ownership

Extracurricular Activities

DEGIS

Volunteer

01.2024 – Present | Schmalkalden, Germany

Rotaract Club of Nashik Nine Hills

Founding Member

2019 – 2020 | Nashik, India

Hobbies/Interests

- Hiking and trekking
- Tinkering with electronics and 3D printing

Autonomous Navigation & Multi-Sensor Fusion on Unitree Go1

Technologies: ROS 1/2, LiDAR, IMU, TF, AMCL, CAD

10.2025 – 12.2025

- Implemented a multi-sensor state estimation pipeline combining LiDAR and IMU to obtain robust pose estimates for a legged robot in indoor environments.
- Established low-latency communication between onboard systems and external computation using UDP/TCP bridges between ROS 1 and ROS 2.
- Evaluated perception and estimation performance experimentally, focusing on reliability and reproducibility under varying sensing conditions.

LLM/NLP Controlled UR5e Cobot System

Tech stack: ROS2, Python, UR5e, MoveIt 2, LangChain, LLM

07.2025 – 11.2025

- Developed a natural language control system for a UR5e collaborative robot, enabling execution of joint and Cartesian motions through conversational commands.
- Implemented the system using ROS2 and the ROSA framework, integrating LLM-based command parsing with a modular tool chain, MoveIt 2 motion planning, and UR driver interfaces for real-time control and feedback.
- Achieved an average response time of approximately 5 seconds from text input to robot execution, while improving operational safety through constraint-aware planning that reduced potential collision scenarios during manipulation tasks.

Design and Implementation of Disaster Response Robot (Bachelor Thesis)

Under K. K. Wagh Institute of Engineering Education and Research

10.2020 – 05.2021

Technologies: Solidworks, YOLO, Sensor Integration, Arduino

- Designed and built an unmanned ground vehicle (UGV) with multi-sensor human detection using pyroelectric and Doppler-based sensors.
- Integrated a vision pipeline using a low-cost ESP32-CAM module and a YOLO-based detector, achieving >88% detection accuracy under varying lighting conditions.
- Conducted experimental testing and system-level validation, balancing sensing, computation, and actuation constraints.